

Advanced Data Mining for Data-Driven Insights and Predictive Modeling

Residency Day 3: Project Deliverable 4

Final Report

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Abstract

This applied project examines how a single retail transaction dataset can support multiple descriptive and predictive data mining tasks when the data are transformed to the correct analytical unit. The UCI Online Retail dataset originally contained 541,909 invoice-line records from a United Kingdom-based online retailer. After removing cancellations, duplicates, missing customer or product information, and non-positive quantity or price values, 392,692 valid transaction rows remained. These records were reorganized into 18,532 invoice-level observations for regression and classification, 4,338 customer profiles for clustering, and 56,880 basket-item records for association rule mining. Exploratory analysis identified strong United Kingdom market concentration, right-skewed revenue, popular household and gift products, and seasonal variation. Random Forest Regression outperformed Linear and Ridge Regression, achieving a test RMSE of 0.265 and an R-squared value of 0.936 on log-transformed invoice revenue. A tuned Random Forest classifier identified high-value invoices with 0.918 accuracy, a 0.824 F1 score, and a 0.968 ROC-AUC. K-Means clustering produced interpretable customer segments, while Apriori rules identified strong relationships among related products. Overall, the findings show that data cleaning, feature engineering, and analytical-unit selection are essential for converting raw retail records into useful decision support. The project supports revenue planning, customer targeting, reactivation, product bundling, and recommendations while emphasizing privacy, fairness, validation, and human oversight.

Introduction

Retail organizations collect large volumes of transactional information through invoices, customer purchases, product returns, and digital sales channels. These records can reveal demand patterns, seasonal behavior, customer value, and relationships among products. However, operational transaction logs are not automatically suitable for analysis. They often contain missing values, duplicate entries, cancellations, extreme values, and variables recorded at different levels of detail. Data mining therefore requires systematic preprocessing, transformation, pattern discovery, model evaluation, and interpretation before the results can support business decisions (Han et al., 2022).

The Online Retail dataset is particularly suitable for an integrated data mining project because it contains real-world purchases and supports several complementary analytical perspectives. The dataset records transactions from December 1, 2010, through December 9, 2011, for a United Kingdom-based non-store retailer that primarily sold all-occasion gift products and served many wholesale customers (Chen, 2015). Previous research using this dataset has also demonstrated that recency, frequency, and monetary value can provide meaningful foundations for customer segmentation and retail strategy (Chen et al., 2012).

The purpose of this project is to develop an end-to-end workflow that converts raw invoice-line data into predictive and descriptive business insights. The analysis has five objectives: examine sales patterns through exploratory data analysis; predict invoice revenue using regression; identify high-value invoices through classification; segment customers using RFM-style behavioral features; and discover product relationships through association rule mining. A central design decision is the creation of separate transaction-level, invoice-level, customer-level, and basket-level datasets because each technique requires a different unit of

analysis. By comparing multiple models, using cross-validation and hyperparameter tuning, and translating results into ethical business recommendations, the project demonstrates how data mining methods can be combined within a practical retail decision-support framework.

Dataset Description and Justification

The Online Retail dataset was selected because it satisfies the project requirements and supports multiple analytical views: transaction lines for EDA, invoice summaries for regression and classification, customer summaries for clustering, and market baskets for association rules. This flexibility makes the dataset suitable for an integrated graduate-level data mining project.

Table 1

Dataset Processing Summary

Stage	Count
Raw dataset rows	541909
Cleaned transaction rows	392692
EDA sample rows	50000
Invoice records	18532
Customer records	4338
Basket transaction-item rows	56880

Note. The raw dataset was cleaned and transformed into derived datasets used across deliverables.

Table 2

Derived Dataset Views and Analytical Purpose

Dataset View	Purpose	Rows
Transaction sample	EDA and revenue/product exploration	50000
Invoice features	Regression and classification	18532
Customer features	RFM-style customer clustering	4338
Basket transactions	Association rule mining	56880

Data Preparation and Cleaning

Data cleaning was necessary because the raw file contained operational artifacts that could distort analysis. Cancelled invoices were removed because they represent reversals rather than completed purchases. Missing CustomerID and Description values were removed because they prevent customer-level analysis and meaningful product interpretation. Duplicate rows were removed to avoid double-counting. Non-positive Quantity and UnitPrice values were removed because they do not represent valid positive sales in this project context.

Feature engineering created TotalPrice at the transaction-line level, invoice-level totals and basket statistics, and customer-level recency, frequency, monetary value, product variety, and average order value. These transformations align the data with specific modeling goals.

Exploratory Data Analysis

EDA showed that the United Kingdom dominates transaction volume and revenue. Product-frequency analysis showed that a relatively small set of gift and household products appears repeatedly across transactions. Revenue analysis showed strong right skew, meaning that a small number of large invoices contribute heavily to sales. This motivated invoice-level aggregation and log transformation for regression modeling.

Figure 1. Monthly revenue trend from Deliverable 1.



Figure 2. Top products by transaction-line frequency.

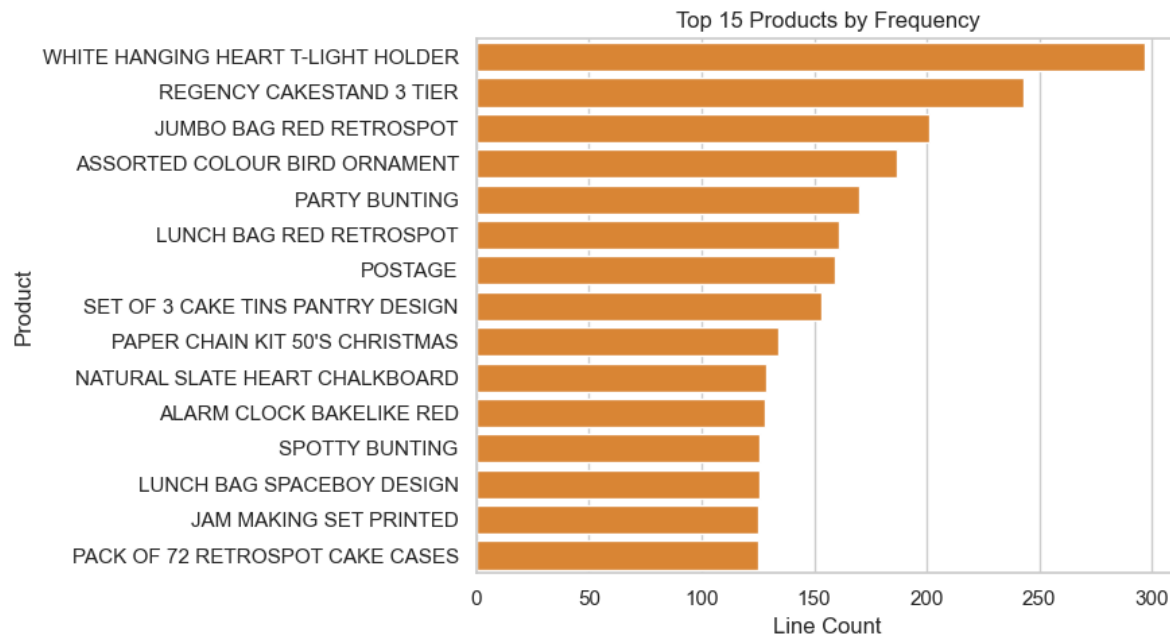
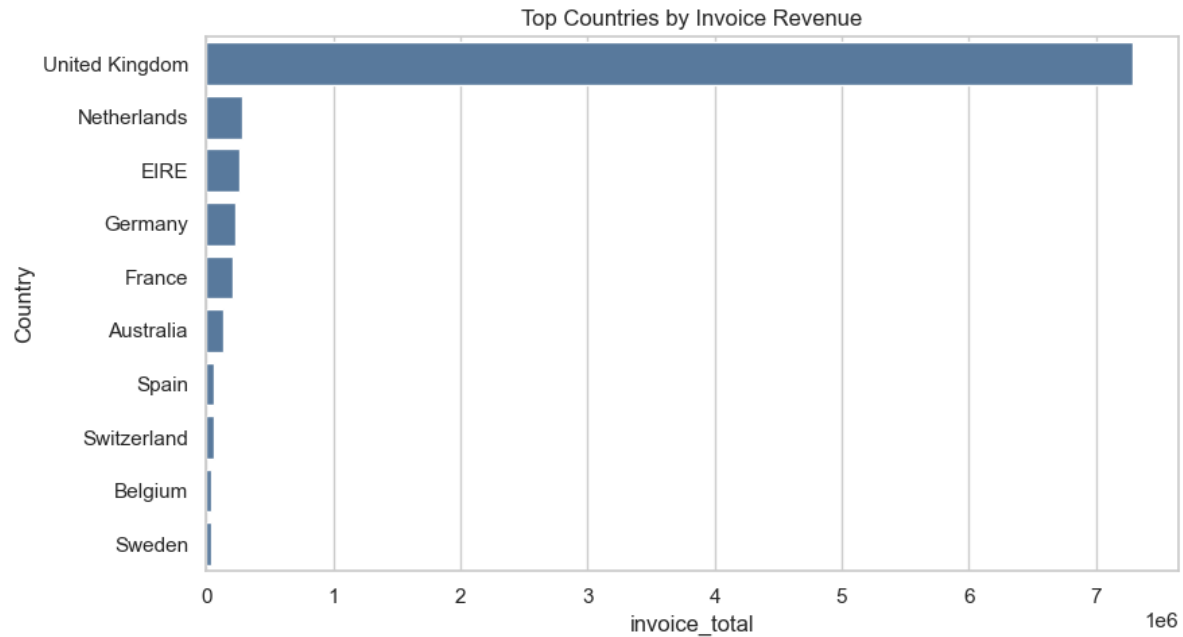


Figure 3. Top countries by invoice revenue.

Regression Modeling and Performance Evaluation

Regression modeling was used to predict invoice revenue. Because invoice totals were highly skewed, the target variable was transformed using $\log(1 + \text{invoice_total})$. Multiple Linear Regression, Ridge Regression, and Random Forest Regression were compared. Numeric predictors were standardized, and Country was one-hot encoded inside a modeling pipeline to prevent preprocessing leakage.

Table 3

Regression Model Performance

Model	Test RMSE	Test R-squared	CV RMSE
Multiple Linear Regression	0.878	0.297	0.909
Ridge Regression	0.879	0.295	0.909
Random Forest Regression	0.265	0.936	0.268

Note. RMSE values are based on the log-transformed invoice total. Lower RMSE and higher R-squared indicate stronger performance.

Figure 4. Regression model comparison by test RMSE.

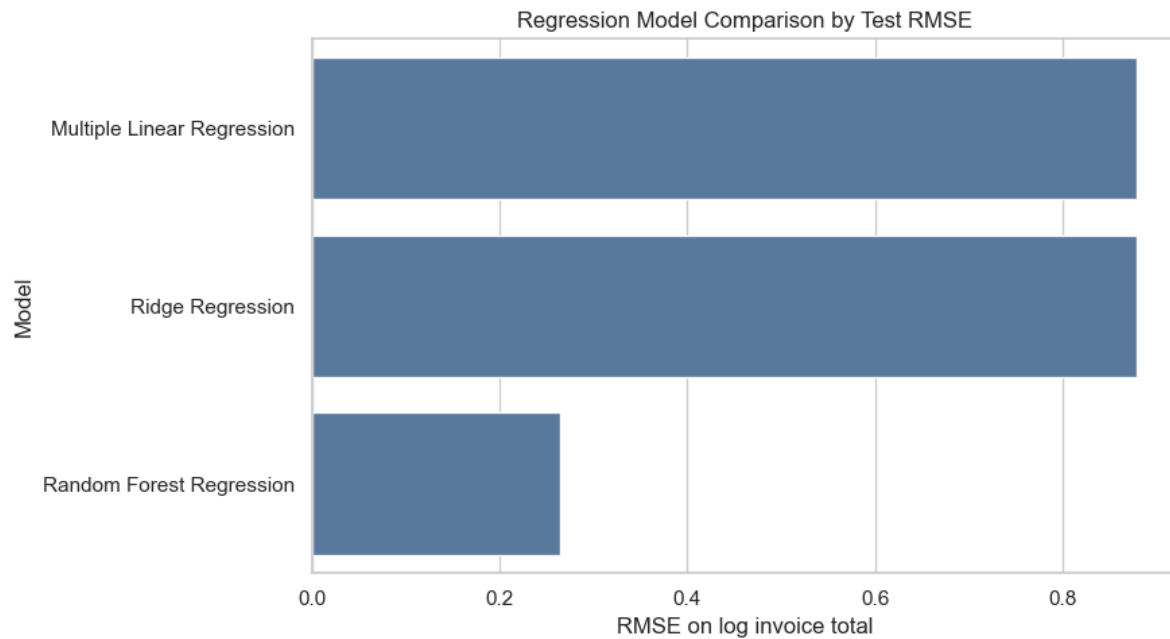
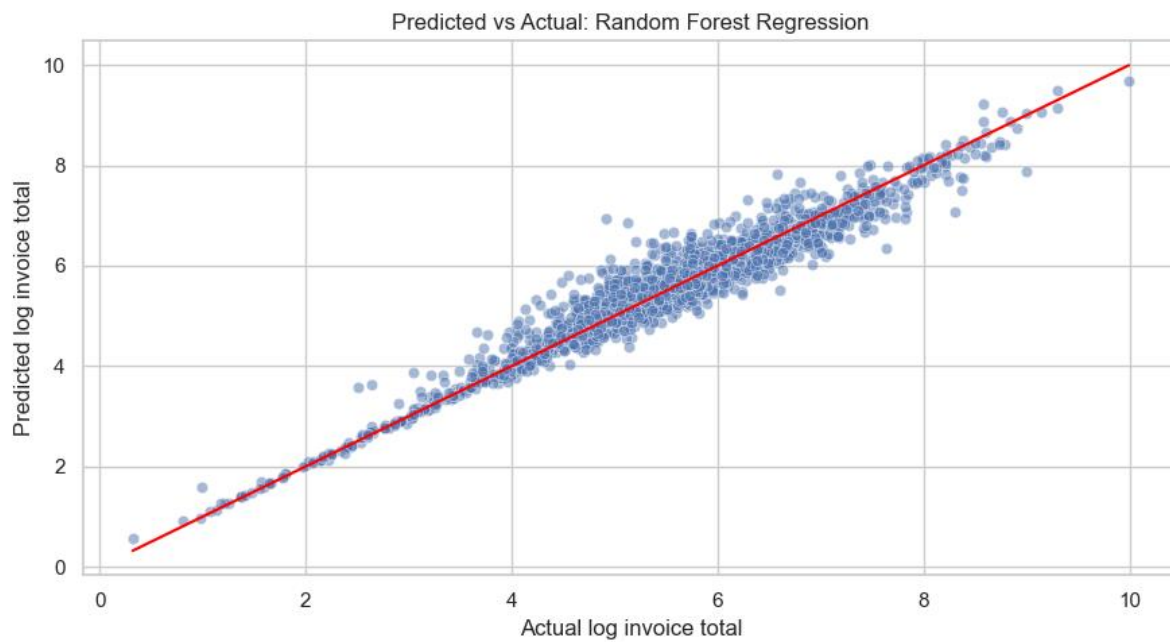


Figure 5. Predicted versus actual values for the best regression model.



The Random Forest Regression model performed best, with a test RMSE of 0.265 and an R-squared value of 0.936. This indicates that nonlinear relationships among quantity, item variety, average price, timing, and country are important for predicting invoice revenue.

Classification Modeling and Hyperparameter Tuning

Classification modeling was used to identify high-value invoices. The high-value target was defined as invoices in the top quartile of invoice totals. Logistic Regression and Random Forest were compared, and Random Forest was tuned with GridSearchCV. Accuracy, F1 score, and ROC-AUC were used because they evaluate different aspects of model performance.

Table 4

Classification Model Performance

Model	Accuracy	F1 Score	ROC-AUC
Logistic Regression	0.874	0.710	0.925
Random Forest	0.914	0.817	0.967
Tuned Random Forest	0.918	0.824	0.968

Figure 6. Confusion matrix for the tuned Random Forest classifier.

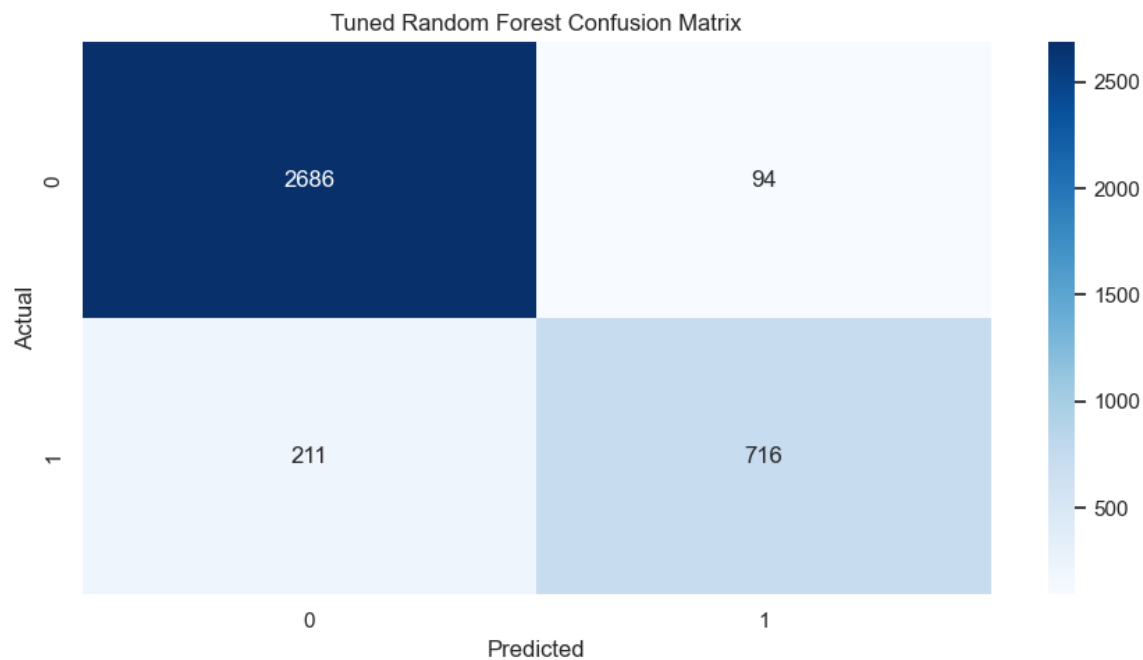
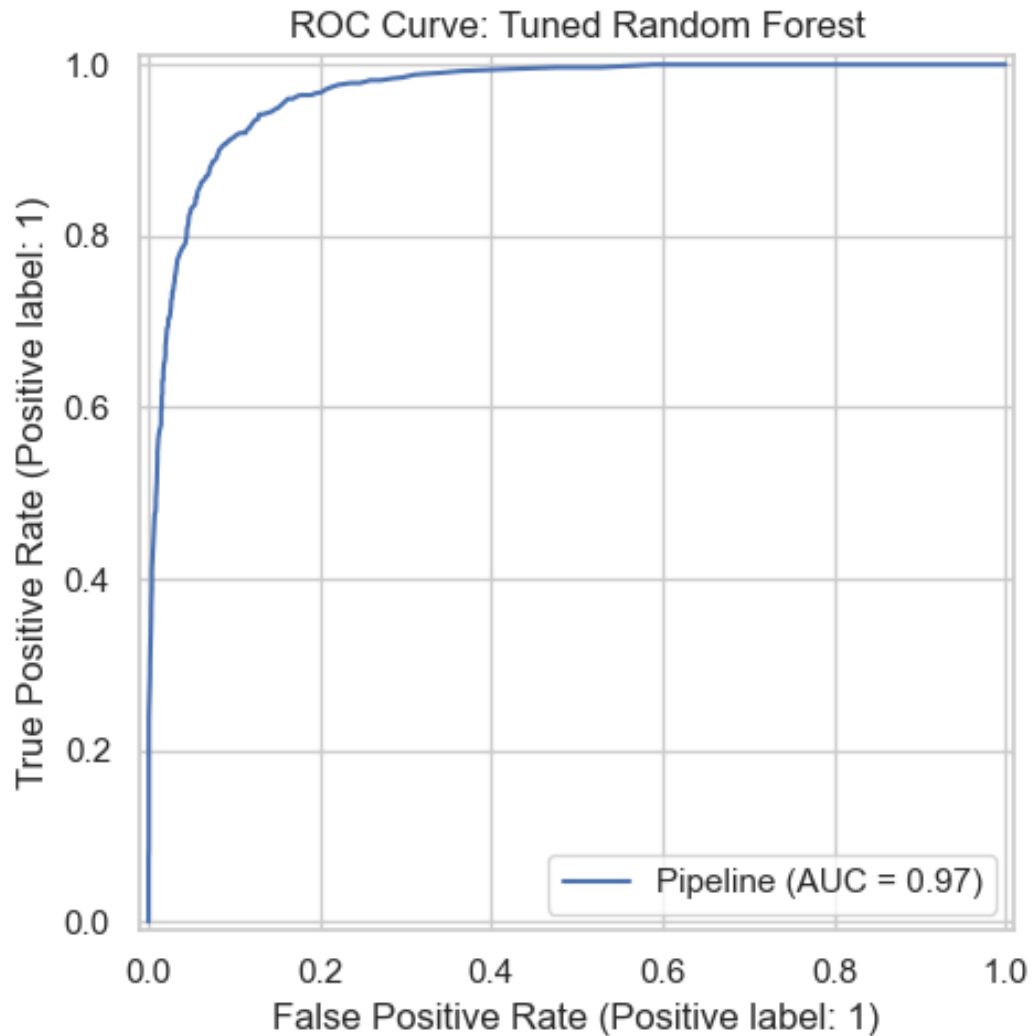


Figure 7. ROC curve for the tuned Random Forest classifier.



The tuned Random Forest produced the strongest classification performance, with accuracy of 0.918, F1 score of 0.824, and ROC-AUC of 0.968. These results suggest that engineered invoice features can effectively identify high-value purchases.

Customer Clustering

Customer clustering was performed using K-Means on RFM-style features: recency, frequency, monetary value, total quantity, unique items, and average order value. These features

summarize how recently customers purchased, how often they purchase, and how much value they generate.

Table 5

Customer Cluster Summary

Cluster	Recency Days	Frequency	Monetary	Unique Items
0	49.130	2.770	859.750	42.830
1	18.280	31.050	33749.780	223.990
2	26.050	9.840	3864.080	164.390
3	250.750	1.480	440.060	22.410

Figure 8. Customer clusters visualized using PCA.



Cluster 1 contains high-frequency, high-monetary customers with broad product variety, making it a strong loyalty or premium-service segment. Cluster 3 has high recency days and low monetary value, suggesting inactive or low-engagement customers who may need reactivation campaigns.

Association Rule Mining

Association rule mining was applied to basket transactions using Apriori. Rules were evaluated with support, confidence, and lift. Support measures how common the full rule is, confidence measures how often the consequent appears when the antecedent appears, and lift measures whether the relationship is stronger than the consequent item's baseline popularity.

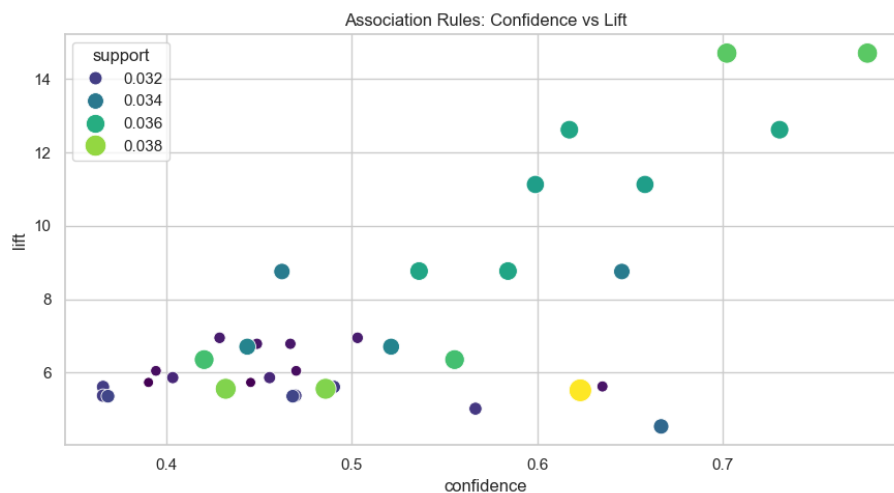
Table 6

Top Association Rules by Lift

Rule	Support	Confidence	Lift
GREEN REGENCY TEACUP AND SAUCER -> ROSES REGENCY TEACUP AND SAUCER	0.037	0.778	14.707
ROSES REGENCY TEACUP AND SAUCER -> GREEN REGENCY TEACUP AND SAUCER	0.037	0.702	14.707
GARDENERS KNEELING PAD CUP OF TEA -> GARDENERS KNEELING PAD KEEP CALM	0.036	0.730	12.621
GARDENERS KNEELING PAD KEEP CALM -> GARDENERS KNEELING PAD CUP OF TEA	0.036	0.617	12.621
ALARM CLOCK BAKELIKE GREEN -> ALARM CLOCK BAKELIKE RED	0.035	0.658	11.128

Note. Rules are ranked by lift and confidence. Long product names are retained because they identify specific retail items.

Figure 9. Association rules plotted by confidence and lift.



The strongest rules involved related product lines, such as Regency teacup-and-saucer items, gardening kneeling pads, and matching alarm clocks. Such rules can support product recommendations, bundle promotions, and website merchandising.

Ethical Considerations

Although the Online Retail dataset does not contain direct identifiers such as customer names, addresses, phone numbers, or email addresses, it includes CustomerID values connected to detailed behavioral purchase histories. Repeated transactions can reveal purchase frequency, spending levels, product preferences, and periods of inactivity. Customer-level records should therefore be treated as sensitive behavioral data rather than as fully anonymous information. Appropriate protections include pseudonymizing or hashing customer identifiers, restricting access according to job responsibilities, encrypting stored data, documenting permitted uses, and retaining only the information required for a legitimate analytical purpose. These practices are consistent with the NIST Privacy Framework, which emphasizes identifying and managing privacy risk throughout the data lifecycle (National Institute of Standards and Technology [NIST], 2020).

Fairness is also important because the high-value invoice classifier and customer clusters could influence how marketing resources, discounts, customer service, or retention efforts are distributed. A retailer may be tempted to prioritize customers who already spend more while reducing attention to lower-spending or less frequent customers. The strong concentration of transactions in the United Kingdom could also cause models to perform less reliably for customers from smaller markets. Before deployment, performance should therefore be examined across relevant customer and geographic groups, and business policies should prevent model scores from becoming the sole basis for service quality or access to opportunities. The NIST AI

Risk Management Framework recommends evaluating fairness, documenting risks, and monitoring model outcomes throughout the system lifecycle (NIST, 2023).

Transparency and human oversight are additional concerns. Random Forest models achieved the strongest predictive performance in this project, but their decision processes are less immediately interpretable than those of Linear or Logistic Regression. Business users should be informed about the purpose of each model, the features used, the meaning of performance metrics, and the possibility of error. Predictions should be treated as estimates based on historical patterns rather than guaranteed outcomes. Periodic review is also necessary because customer behavior, product demand, and market conditions can change, which may reduce model reliability over time (NIST, 2023).

Association rules must also be interpreted carefully. High support, confidence, or lift indicates that products co-occur more often than expected, but it does not prove that purchasing one item causes the purchase of another. Product relationships may be influenced by seasonality, product design, coordinated promotions, or the preferences of a particular customer segment. Association rules are best treated as hypotheses for merchandising or recommendation strategies and should be validated through controlled experiments, A/B testing, customer feedback, or careful post-deployment monitoring (Han et al., 2022).

Practical Recommendations

The retailer should use the invoice-level Random Forest Regression model as a decision-support tool for short-term revenue estimation and operational planning. Because the model captured nonlinear relationships more effectively than Linear and Ridge Regression, it may help estimate invoice value from characteristics such as quantity, item variety, average price, transaction timing, and country. These estimates can support staffing, inventory allocation, sales

monitoring, and the investigation of unusually large or small transactions. The model should be recalibrated regularly and evaluated on newer data before being used in a production environment.

The tuned Random Forest classifier can support the early identification of high-value invoices. When used responsibly, this information may help prioritize personalized follow-up, premium customer service, fraud review, or loyalty outreach. However, the classifier should not be used to lower service quality for customers predicted to generate smaller purchases. A more balanced policy would use high-value predictions to add optional services while maintaining a consistent baseline experience for all customers. Model outcomes and error rates should be monitored after deployment, particularly when the predictions affect customer-facing decisions (NIST, 2023).

The customer clusters should be translated into distinct but flexible marketing strategies. High-frequency, high-monetary customers may be suitable for loyalty benefits, early product access, premium support, or retention incentives. Mid-value customers with broad product interests may respond to product-discovery campaigns and personalized recommendations. Customers with high recency values, meaning they have not purchased recently, may benefit from reactivation messages, time-limited offers, or reminders based on past purchases. RFM-based segmentation is useful because it converts purchase history into interpretable behavioral groups, but the segments should be reviewed over time as customer behavior changes (Chen et al., 2012).

Association rules can support cross-selling, product bundling, recommendation systems, and website merchandising. Products with strong confidence and lift, such as related Regency teacup-and-saucer items, gardening kneeling pads, and matching alarm clocks, could be shown

together on product pages or suggested during checkout. The retailer could also test bundle discounts, coordinated email campaigns, and seasonal displays based on these relationships. Because support, confidence, and lift measure different aspects of co-purchase behavior, rules should be selected using a combination of statistical strength, profitability, inventory availability, and business relevance (Han et al., 2022).

Finally, the retailer should monitor revenue concentration and seasonality. The United Kingdom accounts for most sales in the dataset, creating dependence on a single geographic market. International growth opportunities should be evaluated alongside shipping cost, delivery time, product availability, and local demand. Monthly revenue trends can also inform inventory planning, promotional timing, and temporary staffing. The most effective implementation would combine regression, classification, clustering, and association rules in a monitored analytics dashboard rather than treating any single model as a complete decision system.

Limitations and Future Work

The project is limited by the age, scope, and structure of the source data. The dataset represents one United Kingdom-based online retailer and covers transactions from December 2010 through December 2011 (Chen, 2015). Consequently, the findings may not generalize to current e-commerce environments, other industries, or retailers with different products, customer populations, and geographic markets. Economic conditions, mobile commerce, digital advertising, and customer expectations have changed considerably since the dataset was collected.

Important explanatory and business variables are also missing. The data do not include customer demographics, product category hierarchy, profit margin, shipping cost, marketing exposure, payment method, website browsing behavior, customer satisfaction, or reasons for

returns. Without these attributes, the analysis can identify patterns in completed transactions but cannot fully explain why customers purchased certain products or whether high-revenue transactions were highly profitable. Future datasets should combine transaction records with product, marketing, fulfillment, and customer-experience information while maintaining appropriate privacy controls.

The predictive models also require careful production validation. Several invoice-level features are derived from completed transaction information, so an organization would need to verify that every predictor is available at the exact time a forecast or classification is made. Otherwise, a model may appear accurate during offline testing but be unusable in real-time operations. Future work should define a clear prediction timestamp, prevent target leakage, use temporal train-test splits, assess calibration, and monitor performance drift. Ongoing evaluation is especially important when model outputs affect customer-facing decisions (NIST, 2023).

The clustering and association analyses depend on methodological choices. K-Means results are influenced by feature selection, scaling, outliers, and the chosen number of clusters. Alternative methods such as hierarchical clustering, DBSCAN, or Gaussian mixture models could be compared to test segment stability. Association rules are similarly affected by support and confidence thresholds; low thresholds may create many weak rules, while high thresholds may remove rare but valuable combinations. Future analysis could rank rules by profit, season, customer segment, and product category in addition to support, confidence, and lift (Han et al., 2022).

Future work could extend the project through time-series forecasting, customer lifetime value modeling, product taxonomy enrichment, churn prediction, uplift modeling, and recommendation systems. Controlled A/B testing would be especially valuable for determining

whether bundles, personalized offers, or reactivation campaigns create incremental improvements rather than merely identifying customers who would have purchased anyway. These extensions would move the project from descriptive pattern discovery toward measurable, continuously monitored business impact.

Conclusion

This project demonstrates that a single retail transaction dataset can support a complete data mining workflow when the data are carefully cleaned and reorganized around specific business questions. Raw invoice-line records were transformed into transaction-level, invoice-level, customer-level, and basket-level views. This separation was essential because exploratory analysis, predictive modeling, customer segmentation, and association rule mining each require a different analytical unit.

The results also show the value of feature engineering and model comparison. Random Forest Regression substantially outperformed the linear alternatives for predicting log-transformed invoice revenue, indicating that invoice value depends on nonlinear interactions among the available features. The tuned Random Forest classifier also identified high-value invoices with strong accuracy, F1, and ROC-AUC results. At the descriptive level, K-Means produced actionable customer segments, and Apriori identified product relationships that can support cross-selling and merchandising.

From a business perspective, the combined workflow supports revenue planning, customer prioritization, loyalty and reactivation campaigns, product bundling, and website recommendations. At the same time, the results should not be treated as automatic decisions. Privacy protections, fairness analysis, human review, model monitoring, and controlled testing are necessary to ensure that analytical recommendations remain reliable and responsible. This

aligns with the broader principle that useful data mining depends not only on algorithm selection but also on preprocessing quality, appropriate evaluation, contextual interpretation, and governance (Han et al., 2022; NIST, 2023).

Overall, the strongest lesson from the project is that analytical quality begins with matching the data structure and engineered features to the business objective. When this alignment is achieved, even one transactional dataset can provide complementary insights about forecasting, customer behavior, and product relationships. The project therefore offers a practical foundation for a larger retail analytics system that can be expanded with newer data, profitability measures, customer lifetime value, time-series forecasting, and experimental validation.

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